



FINTECH · APPLIED AI · INVESTMENT RESEARCH

# SentiVest AI

*Turn financial noise into investment insights.*

RAG-Powered Stock Sentiment Intelligence Platform

Business Case & Solution Overview

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# Industry

*The business domain SentiVest AI operates in*



## FinTech — Investment Research & Capital Markets Intelligence

Part of the broader Wealth-Tech and decision-support software category, serving retail investors, buy-side/sell-side analysts, brokerages, and asset managers who need to interpret market sentiment quickly.



### Always-On

Markets and news never stop moving



### Multi-Source

News, filings, calls, analyst notes



### Two Audiences

Retail investors and professional analysts

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# Company Examples

*Organizations facing similar challenges*



## Bloomberg Terminal

Deep data access, but manual synthesis across sources



## Refinitiv / LSEG

Enterprise-grade feeds, limited grounded Q&A



## AlphaSense

Strong document search, less finance-tuned sentiment scoring



## Retail Broking Apps

Real-time prices, minimal narrative context



## Robo-Advisors

Portfolio automation without evidence-backed insight



## In-House Research Desks

High cost, slow manual report turnaround

## 3 Business Background

*Why sentiment analysis has become mission-critical*

Retail investing has surged over the past decade, and both retail and institutional investors now face an unprecedented volume of financial content — news wires, exchange filings, earnings calls, and analyst notes — published continuously across fragmented channels.

Sentiment has become a leading input to investment decisions — but synthesizing it manually across sources is slow, inconsistent, and doesn't scale with how fast the market moves.

### WHERE THE NOISE COMES FROM



**News & Media**



**Regulatory Filings**



**Earnings Calls**



**Analyst Reports**

## 4 Business Problem

*The core pain point*



**Investors cannot efficiently convert fragmented, high-volume financial information into a clear, trustworthy view of market sentiment for a given stock.**



Information overload across disconnected, high-volume sources



No single grounded answer to 'why is sentiment shifting?'



Manual synthesis doesn't scale with news velocity

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## Why Existing Solutions Fail

*Current approaches leave a gap*



### Generic LLM Chatbots

Answer from pretrained knowledge  
— no live evidence, prone to  
hallucination



### Generic Sentiment Models

Miss financial phrasing (e.g. 'guided  
down' read as neutral)



### Single-Source Aggregators

One source at a time — no cross-  
source synthesis



### SentiVest AI

Multi-source RAG + finance-tuned  
scoring, fully cited

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## Business Impact

*Operational, customer, and financial consequences*



### OPERATIONAL

Analyst and investor hours lost to manual cross-source reading and reconciliation



### CUSTOMER

Trust erodes when AI-generated answers can't be traced back to real evidence



### FINANCIAL

Delayed or misinformed decisions carry real downside and missed-opportunity risk

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## Problem Statement



*Build an evidence-grounded, RAG-powered platform that synthesizes multi-source financial sentiment into transparent, source-cited insight — in seconds, not hours.*





# Business Objectives

*What success looks like*



Reduce manual research time per sentiment query from hours to seconds



Ground every insight in retrievable, citable evidence



Apply finance-tuned sentiment scoring — not generic NLP



Provide a self-serve dashboard and natural-language Q&A for any tracked stock



Support both retail-investor and professional-analyst research workflows

# 9 Functional Requirements

*Core system capabilities*



## Multi-Source Ingestion

News, filings, earnings calls, analyst reports



## Semantic Retrieval

Vector search over embedded document chunks



## Sentiment Scoring

Finance-tuned FinBERT classification



## Natural-Language Q&A

Ask questions about any tracked stock



## Trend Dashboard

Historical sentiment visualization



## Source Citations

Every claim linked to its evidence

# 10 Non-Functional Requirements

*Performance, security, and scalability*



## Performance

Sub-few-second retrieval and report generation per query



## Security

API-key gated ingestion; secrets never exposed to clients



## Scalability

Pluggable vector store and async task queue for growth



## Reliability & Auditability

Every claim traceable to a specific cited source

# 11 Data Required

*Datasets, APIs, and data sources*



## Financial News APIs

Real-time and historical company news coverage



## Regulatory Filings

SEC EDGAR full-text search / exchange announcements



## Earnings Call Transcripts

Management commentary and Q&A tone



## Analyst Reports

Third-party research and rating commentary



## Company & Ticker Metadata

Fundamentals, sector, exchange reference data

# 12 Suggested Tech Stack

*Languages, frameworks, databases, and tools*



## Frontend

- Next.js 14 (App Router)
- TypeScript
- Tailwind CSS
- Recharts



## Backend

- FastAPI (async Python)
- SQLAlchemy 2.0
- PostgreSQL
- Alembic migrations



## AI & RAG

- OpenAI LLM + Embeddings
- FinBERT sentiment model
- Chroma / Pinecone / Qdrant



## Infra

- Docker Compose
- Celery + Redis
- Nightly scheduled ingestion

# 13 AI / ML Concepts

*The techniques behind the platform*



## Semantic Embeddings & Vector Search

Represent text as vectors; retrieve by similarity, not keywords



## Retrieval-Augmented Generation (RAG)

Ground LLM output in retrieved documents, not pretrained memory



## Finance-Tuned Sentiment Classification

FinBERT correctly reads financial phrasing general models miss



## LLM-Based Grounded Summarization

Synthesize a narrative strictly from cited source excerpts



## Evidence Aggregation & Confidence Scoring

Combine per-document scores into an auditable overall signal

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# KPIs

*Target success metrics for the platform*

**< 5 sec**

Target report generation latency  
per query

**100%**

Claims backed by a cited,  
retrievable source

**Daily**

Ingestion refresh cadence per  
tracked stock

**Multi-Source**

News + filings + earnings + analyst  
coverage per report

*Targets guiding design and evaluation — to be measured against production usage.*

# 15 Business Outcome

*Expected benefits*

Faster, evidence-backed research turns raw financial noise into decision-ready insight — reducing manual synthesis time, increasing trust through full source traceability, and giving investors a defensible, auditable view of market sentiment.



## Faster Research Cycles

Minutes of AI-assisted synthesis instead of hours of manual reading



## Explainable, Auditable Output

Every insight traces back to a real, citable source



## Differentiated Research Product

A defensible foundation for a premium investor-research offering



# 16 Expected Deliverables

*Final project artifacts*



FastAPI backend with a full RAG sentiment-analysis pipeline



PostgreSQL + pluggable vector-store data layer



Next.js investor dashboard, stock detail view, and chat Q&A interface



Docker Compose deployment for backend, frontend, and workers



FinBERT-based finance-tuned sentiment scoring service



API reference and architecture documentation



Multi-source ingestion pipeline — news, filings, earnings, analyst reports



# Thank You

*Turn financial noise into investment insights.*

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